HROC · NCAN · DEEP-LEARNING TRACK

Six animals, both directions, and a positive result.

Every control you asked for is now built and run.

Suchith Rao • for Dr. Carp • August 2026

YOUR ASKS, LAST MEETING

All four are built and run

1	**Randomised-order control**	Shuffle trial order, rerun the drift. If it still drifts, it's baked into the numbers.	**PASSES — null ≈ 0 in every animal**
2	**Pre-stimulus background**	Motoneuron-pool excitability, 20 ms before the stimulus, from fragment 0.	**Built — now a covariate everywhere**
3	**Last 10 baseline days**	Early baseline had settings changes, so only the last 10 days are the reference.	**Adopted — visibly cleaner**
4	**A strong up-conditioned animal**	Animal 12 was weak. Scanned every log for reward-criterion progression.	**Found animals 3 and 4**

THE DATASET

Six animals, both directions

6

animals processed

2.7M

trials decoded

3 / 3

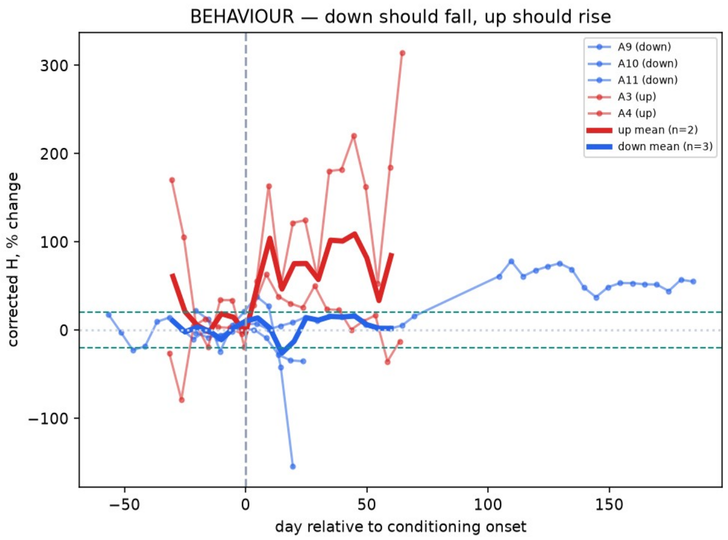
down / up

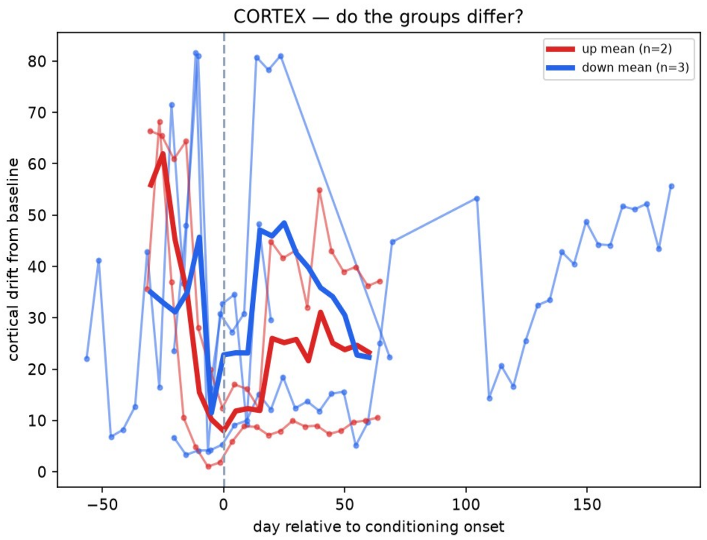
How we found the up animals

We scanned every animal's log without downloading the data. The reward criterion (RW) is a success readout — you raise the bar when the rat beats it, lower it when it struggles.

Animal 4: RW 150→220 (+47%) • Animal 3: RW 90→190 (+111%)   Animals 1 and 6: criterion lowered — failed

Conditioning works in both directions





Left panel

Corrected H, aligned to each animal's own onset. Red (up) rises, blue (down) falls. Group difference +72 percentage points.

Animals 9, 11 (down) and 3 (up) clear your 20% criterion.

All with stimulus and pre-stimulus background regressed out.

RESULT 2 · THE DRIFT QUESTION, ANSWERED

Cortical drift is not conditioning

The logic

Recording drift doesn't care which direction you trained the animal. Conditioning does. So if the drift is learning, it must differ between up and down.

It doesn't

Up 21.9 vs down 34.6, $t = -1.01$, not significant. No direction dependence, so the drift is recording nonstationarity.

Sham null (baseline split)	**fails in 5 of 6 animals**
Randomised-order null	**passes — ≈0 everywhere**
Direction contrast	**no difference ($t = -1.01$)**

Three independent controls agree. This question is answered.

SO WE CHANGED THE QUESTION

A question drift can't contaminate

Old question

"Did the average cortical state move over weeks?"

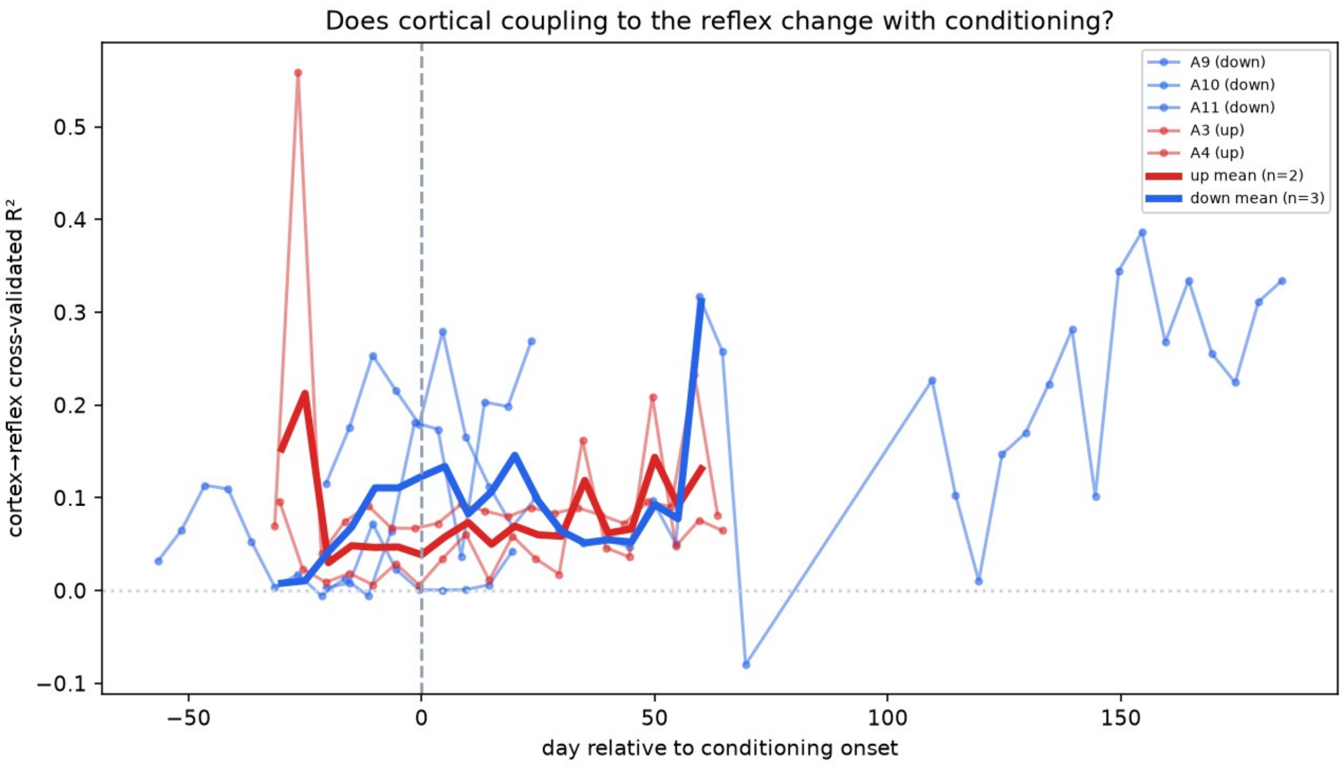
New question

"Can cortex predict the reflex trial by trial?"

1. **Train and test inside one 5-day block** — Slow drift moves train and test together, so it cancels. Drift cannot create within-block predictive structure.
2. **Stimulus and background removed first** — So the decoder can't cheat by reading stimulus size or muscle tone.
3. **Within-block label shuffle as the null** — Real coupling has to beat a shuffle of its own block.

RESULT 3 · THE POSITIVE FINDING

Cortex predicts the reflex, trial by trial



R^2 up to 0.35

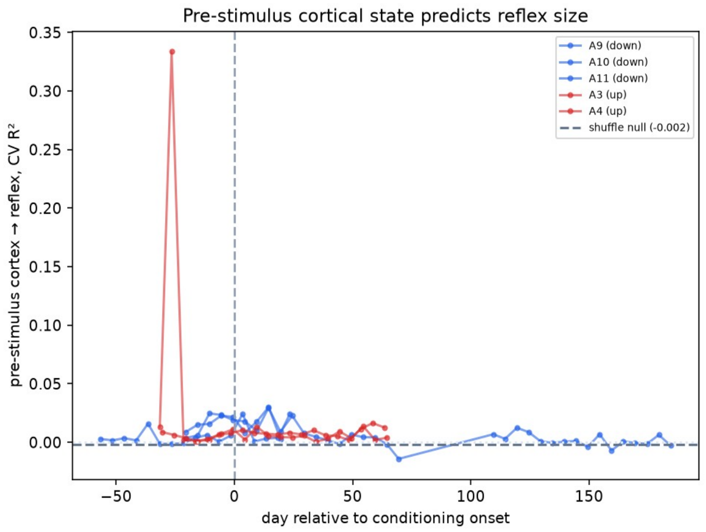
against a shuffle null of ≈ 0. Present in every animal (0.02–0.19 mean).

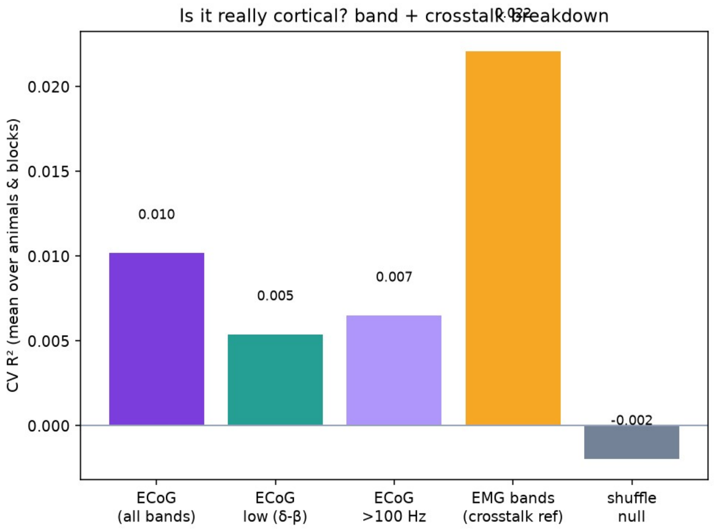
Cortical activity carries trial-by-trial information about the reflex that isn't stimulus intensity, isn't muscle tone, and isn't drift.

Whether it CHANGES with conditioning is not resolved (t = 0.69, n = 5).

RESULT 4 · IS IT REALLY CORTICAL?

Pre-stimulus state and the crosstalk check





What we tested

Cortical band power measured entirely BEFORE the stimulus — it can't contain a response that hasn't happened.

It works, but weakly: R^2 0.010 vs null −0.002.

And pre-stimulus MUSCLE predicts better (0.022) — exactly what you predicted about motoneuron-pool excitability. A good check that the measurement is right.

WHERE WE STAND

Three findings, honestly rated

SOLID

Behaviour, bidirectional

Conditioning works both directions, fully controlled for stimulus and background. Clears your 20% criterion in 3 animals.

REAL

Cortex→reflex coupling

R^2 up to 0.35 over a ≈0 null, drift-immune by design, present in every animal. This is the positive result.

ANSWERED: NO

Drift = learning?

Three independent controls agree the average-state drift is recording nonstationarity. Useful negative.

LAST FEW WEEKS

Plan for the rest of the project

1. **Add the remaining down animals** — 7, 8, 13, 14, 15 are logged and ready. n=5→10 is what resolves whether coupling changes with conditioning.
2. **Finish the crosstalk test** — Spectral/temporal dissociation for the post-stimulus coupling. The one thing a reviewer will demand.
3. **Frequency-resolved coupling** — Which bands carry the reflex information? Sharpens the result and the crosstalk argument.
4. **Write it up** — Behaviour + coupling + the controls. Draft while the last animals process.

The paper: bidirectional conditioning + a drift-immune cortical coupling result + the control methodology.

WHERE I NEED YOUR READ

Four questions

1 **Is the coupling result interesting to you?**

Cortex predicts reflex size trial by trial, independent of stimulus and background. Is that a finding worth building the paper on?

2 **How would you rule out crosstalk?**

Both channels record simultaneously. What would convince you the ECoG isn't picking up muscle?

3 **Which animals next?**

7, 8, 13, 14, 15 are all down. Any more up-conditioned animals worth finding?

4 **Is the negative worth reporting?**

Three controls say the average-state drift is recording, not learning. Worth publishing as a caution?

All code, figures and per-animal results are in the repo.